

Predicting Student Exam Scores Using Machine Learning: Analyzing Key Performance Factors

Final Project AOL Machine Learning

Group 6 Data Science:

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INTRODUCTION

- **Problem Statement:** The process of understanding and predicting student academic performance that analyzes various factors that can affect test results. This model is designed to provide accurate insight into the factors that most affect academic achievement.
- **Objectives:** Analyze the factors that have a major impact on student performance, so that they can be used as a reference in improving learning programs.
- **Data Description:** The StudentPerformanceFactors.csv dataset consists of 6,607 student data, covering various aspects. Attributes are used as predictors in determining the target variable namely student test scores.

5. **Correlation Analysis:** A correlation matrix is generated to identify features most strongly related to the target variable: **Exam Score**. The most relevant features, such as **Hours Studied** and **Attendance**, are selected for the model.

DATA UNDERSTANDING

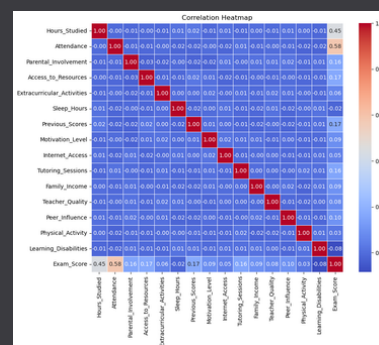
In this stage, we understand the data set to gain insights into its structure and characteristics. The data set contains 6,607 student records, which contain both numeric and categorical data. The main steps in this stage include:

1. **Data Structure Overview:** Check the size of the data set, column names and data types.
2. **Summary Statistics:** Provide summary statistics for numeric features which can help us understand the distribution of data.
3. **Missing Values:** Identifies columns with missing data, which guides the next steps for data cleaning.

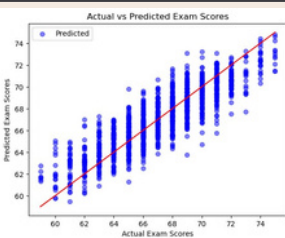
This phase includes cleaning, transforming, and organizing the data to ensure it's ready for machine learning. Key steps are:

1. **Handling Missing Values:** Rows with missing values in crucial columns are removed.
2. **Categorical and Numerical Feature Division:** Features are divided into categorical and numerical types.
3. **Mapping Categorical Values:** Categorical columns are mapped to numeric values using predefined mappings.
4. **Outlier Detection and Removal:** Outliers in critical variables are detected using the IQR method, and removed to prevent bias in the model.

DATA PREPARATION



MACHINE LEARNING MODELING



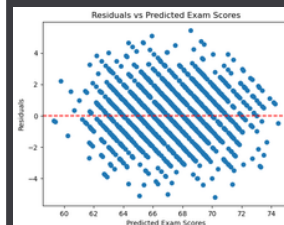
There is **linear relationship** between Hours Studied and Exam Score, as well as Attendance and Exam Score, leading us to choose a **linear regression model** for prediction. The process includes:

- **Data Split:** The data is divided into an 80% training set and a 20% testing set using train_test_split.
- **Model Creation and Training:** A Linear Regression model is trained on the training data, using Attendance and Hours Studied as the features.
- **Prediction:** The model is used to predict exam scores for the test set, and the predicted results are compared with the actual scores.

The Predicted vs Actual plot shows that the predicted scores closely align with the actual scores, with a red line representing an ideal prediction.

ANALYSIS

The Residual plot reveals no significant pattern, suggesting that the model captures the relationship between predictors and the target **effectively without bias**.



EVALUATION

The model was evaluated using the following metrics:

1. **Mean Squared Error (MSE):** The MSE was 2.94, indicating that the predicted scores are close to the actual scores.
2. **R-squared:** With a value of 0.72, the model explains 72.0% of the variance in exam scores.
3. **Root Mean Squared Error (RMSE):** The RMSE of 1.72 shows that predicted scores are on average 1.72 points off from the actual scores.
4. **Mean Absolute Error (MAE):** The MAE of 1.38 indicates an average deviation of 1.38 points between the predicted and actual scores.

CONCLUSION

This project aims to predict the **exam score** of students based on some specific key factors like study habits and attendance, using **machine learning**. Following data cleaning and preprocessing steps on the dataset, we have developed a **Linear Regression model** to predict the student's exam scores based on **Study Hours** and **Attendance**. The analysis revealed a **linear relationship** between predictors and the target variable, therefore, the choice of the **linear regression** approach is appropriate.

The model showed **promising results** with an R-squared of 0.72, which means that almost 72.0% of the variance in exam scores was explained by the model. Also, the Mean Squared Error of 2.94, Root Mean Squared Error of 1.72, and Mean Absolute Error of 1.38 indicate that the **predictions of the model are pretty accurate**.

Therefore, this project informs educators and policymakers on how best to improve student learning outcomes by understanding the factors that influence academic performance and developing a predictive model. Further refinement of the model and **inclusion of more variables** can result in **higher accuracy** of predictions and **deeper insights** into student performance.